AUTOMATED PAYMENT REMINDER SYSTEM USING ZAPIER

Workflow Automation Project using Zapier Agents, Google Sheets and Gmail

Project Report

Submitted by

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1. Executive Summary

This project presents an automated payment reminder system developed using a Zapier Agent connected to Google Sheets, Google Drive and Gmail. The purpose of the system is to reduce the repetitive manual work involved in identifying pending invoices and sending payment reminders to the correct recipients.

The agent is configured with natural-language instructions and a defined set of tools. It locates the invoice spreadsheet P-0 in Google Drive, reads all invoice records, and processes only those marked as Pending. Each eligible invoice is routed to a recipient based on its Type: Maintenance invoices are sent to the configured A1 address and Project invoices to the configured A2 address. A personalized email is generated from the spreadsheet fields, and after it is sent, the corresponding row is updated to record that the first reminder has been issued.

The project demonstrates how no-code, AI-assisted automation can be applied to a practical finance process, and it establishes a foundation for payment confirmation through email replies, transaction ID management, a second reminder after 24 hours, and escalation of overdue payments.

2. Problem Statement

Payment follow-up is often handled manually: someone reviews invoice records, identifies outstanding payments, works out who should be contacted, prepares reminder messages and keeps a record of what was sent. When this is repeated every day, it consumes administrative time and increases the chance of missed, duplicated or inconsistent follow-ups.

This project addresses the problem with an automation that connects invoice data directly to email communication. Instead of checking each invoice by hand, the agent evaluates every record, decides whether a reminder is due and to whom it should go, sends it, and updates the record.

3. Project Objectives

- Automate the identification of pending payment records from the P-0 spreadsheet.
- Run the workflow on a daily schedule (9:00 AM) and on demand for testing.
- Apply conditional logic so that only invoices with Status = Pending are processed.
- Route each reminder by invoice Type: Maintenance to A1 and Project to A2.
- Generate personalized payment reminder emails using dynamic spreadsheet fields.
- Update only the relevant cells of the processed row after a reminder is sent.
- Track reminder activity through the Reminder Count and First Reminder Timestamp fields.
- Provide a scalable foundation for payment confirmation, transaction ID management, second reminders and escalation.

4. Tools and Technologies Used

Tool	Role in Project
Zapier Agents	AI agent platform used to define the workflow in natural-language instructions and connect the spreadsheet, Drive and Gmail tools.
Google Sheets	Stores the invoice records in the spreadsheet P-0 (Sheet1). Used to read rows (Get Many Spreadsheet Rows – Advanced) and update rows (Update Spreadsheet Row). Also added as a knowledge source for the agent.
Google Drive	Locates the P-0 spreadsheet (Find a File) in the cloud before the rows are read.

Tool	Role in Project
Gmail	Sends the personalized payment reminders (Send Email). Find Email and Create Draft are also connected to support reply monitoring and review.
Agent Preview	Zapier's testing panel, used to run the agent on demand and approve each action during testing.

5. System Workflow

The agent follows a sequence of trigger, file lookup, data retrieval, conditional processing, recipient routing, email communication and record update.

Step	Zapier Component	Function
1	Trigger – Schedule / On Demand	Runs every day at 9:00 AM; started on demand from Agent Preview during testing.
2	Google Drive – Find a File	Locates the spreadsheet named P-0.
3	Google Sheets – Get Many Spreadsheet Rows (Advanced)	Reads all invoice records from Sheet1 of P-0.
4	Agent logic – Status filter	Keeps only rows where Status is exactly Pending; all other rows are skipped.
5	Agent logic – Type routing	Maintenance → A1 address; Project → A2 address. Unsupported types or missing recipients are skipped.
6	Gmail – Send Email	Sends a personalized payment reminder using the invoice fields.
7	Google Sheets – Update Spreadsheet Row	Sets Reminder Count to 1 and records the First Reminder Timestamp (Overwrite All Columns = False).

Workflow Summary: Trigger → Find P-0 → Retrieve Invoice Rows → Filter Pending → Route by Type → Send Reminder → Update Spreadsheet

6. Input Data Structure

The P-0 spreadsheet is the primary data source. Its fields represent invoice details, payment status, reminder tracking and recipient information.

Field	Purpose
Sr no.	Serial/reference number
Invoice ID	Unique identifier of the invoice; used to target the correct row for updates
Type	Invoice type (Maintenance or Project); decides the recipient
Name	Name used to personalize the email
Amount	Invoice/payment amount (₹)
Account No	Bank account reference
Bank Name	Bank associated with the record
Status	Current payment status (Pending, Paid, Completed)
Transaction ID	Transaction reference, when verified
Reminder Count	Number of reminders already issued

Field	Purpose
Escalated	Indicates whether the record has been escalated
Email	Designated test recipient address
First Reminder Timestamp	Time the first reminder was sent (an email log is used if the column is absent)

Sample records used for testing:

Invoice ID	Type	Name	Amount	Bank	Status	Txn ID	Reminder
1001	Maintenance	Anirudha	45000	SBI	Paid	111	1
1002	Project	Gautham	28000	ICICI	Pending	222	1
1003	Project	Rohit	32000	HDFC	Pending	333	
1004	Maintenance	Aditya	35000	SBI	Pending	444	
1005	Project	Hitarth	24000	ICICI	Pending	555	

7. Conditional Logic

The first-reminder workflow uses the following eligibility and routing rules:

Status = Pending AND Type ∈ {Maintenance, Project} AND recipient configured

Type	Recipient
Maintenance	Configured A1 email address
Project	Configured A2 email address
Any other type	Skipped

If all conditions are satisfied, the record continues to the Gmail action. If the Status is Paid, Completed or any other non-pending value, the record is skipped. A Pending record with an unsupported Type or no configured recipient is also skipped. Emails are sent only to explicitly authorized recipients, which prevents unintended or incomplete email actions.

8. Payment Reminder Email

The email uses dynamic fields from the spreadsheet so that each eligible invoice receives a personalized message.

Subject: Payment Reminder – Invoice [Invoice ID] Due

> Dear [Name],
>
> This is a friendly reminder that payment for Invoice ID [Invoice ID], amounting to ₹[Amount], is currently pending.
>
> Please find the payment details below:
>
> Invoice ID: [Invoice ID]
> Invoice Type: [Type]
> Pending Amount: ₹[Amount]
> Account Number: [Account No]
> Bank Name: [Bank Name]
> Transaction ID: [Transaction ID, if available]
>
> Kindly complete the payment at the earliest to avoid any inconvenience. If you have already completed the payment, please reply to this email with your payment confirmation.
>
> Thank you for your cooperation.
>
> Regards,
> Payments Team

9. Spreadsheet Update and Tracking

Once the reminder email is sent successfully, the agent updates the matching row in P-0, located by Invoice ID. Reminder Count changes from 0 (blank) to 1, and the First Reminder Timestamp is recorded. The Update Spreadsheet Row action runs with Overwrite All Columns set to False, so only the intended cells change and all other values and formatting are preserved. The Status field is not changed merely because a reminder was sent; it remains Pending until payment is confirmed. If the email fails to send, the Reminder Count is not updated.

Stage	Reminder Tracking	Status
Before reminder	Reminder Count = 0 / blank	Pending
After first reminder	Reminder Count = 1, timestamp recorded	Pending
After payment confirmation	Future workflow (V1)	Completed

10. Testing and Results

The agent was tested in Zapier's Agent Preview using the P-0 test data, with each action reviewed and approved before it ran. Records with a Pending status and a supported Type are expected to trigger the Gmail action; all other records are expected to be skipped.

Scenario	Expected Action
Status = Pending + Type = Maintenance	Send personalized reminder to A1
Status = Pending + Type = Project	Send personalized reminder to A2

Scenario	Expected Action
Status = Paid	Skip
Status = Completed	Skip
Status = Pending + unsupported Type / no recipient	Skip
Successful first reminder	Update Reminder Count to 1 and record timestamp

Observed result: The reminder for Invoice 1001 (Anirudha, ₹45,000, SBI, Maintenance) was delivered to the designated inbox with all dynamic fields correctly populated (Figure 3). The Reminder column in P-0 shows a value of 1 for the processed rows, confirming that the spreadsheet update step worked (Figure 4).

11. Future Enhancements

V1 – Payment Confirmation: Monitor the Gmail inbox for replies to reminder emails using Gmail: Find Email. When an authorized reply clearly confirms payment, update the invoice Status from Pending to Completed and log the confirmation time. Ambiguous replies leave the Status unchanged.

V2 – Transaction ID Management: Update the Transaction ID cell only when a verified reference is available, never generating or reusing IDs.

V3 – Second Reminder after 24 Hours: Send an HTML second reminder, with the heading and escalation warning highlighted in red, when Status = Pending, Reminder Count = 1, Transaction ID exists, Escalated is blank or No, and at least 24 hours have passed since the first reminder. After sending, update Reminder Count to 2; never send it twice.

Escalation: If payment remains pending after the reminder stages, notify the concerned authority and mark the record as escalated.

Audit Log and Dashboard: Maintain a log of every reminder, status change and error, and summarize pending invoices, reminders sent, completed payments and escalated records in a dashboard.

Publishing: Publish the agent with the 9:00 AM schedule trigger so it runs without manual approval.

12. Conclusion

The Automated Payment Reminder System shows how a repetitive finance-related administrative process can be converted into a structured, AI-assisted no-code workflow. By combining scheduled execution, spreadsheet-based data retrieval, conditional filtering, type-based recipient routing, dynamic email generation and targeted spreadsheet updates, the solution reduces manual effort while keeping a clear record of what has been sent.

The current implementation covers the first payment reminder and provides a clear foundation for more advanced automation. The planned payment confirmation, transaction ID management, second reminder and escalation features can extend the workflow into a complete payment follow-up lifecycle.

13. Project Evidence and Submission Files

The project is documented using the following supporting files:

- **prompt.md** – full agent instructions covering V0 (first reminder) to V3 (second reminder), logging and spreadsheet-integrity rules.
- **P-0 Google Sheet** – sample invoice dataset used for demonstration.

- **Zapier screenshots** – evidence of the configured agent, its tools, the delivered email and the updated sheet.
- **This report** – structured documentation of the project.

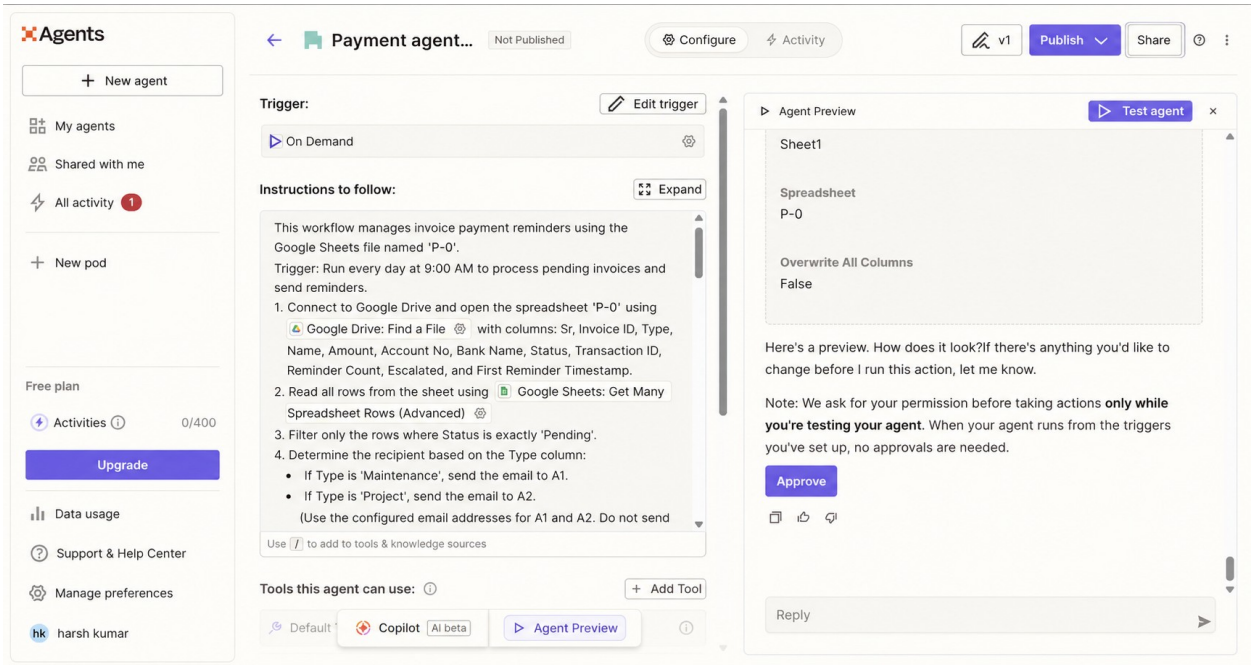


Figure 1: Zapier Agent configuration – trigger, instructions and Agent Preview

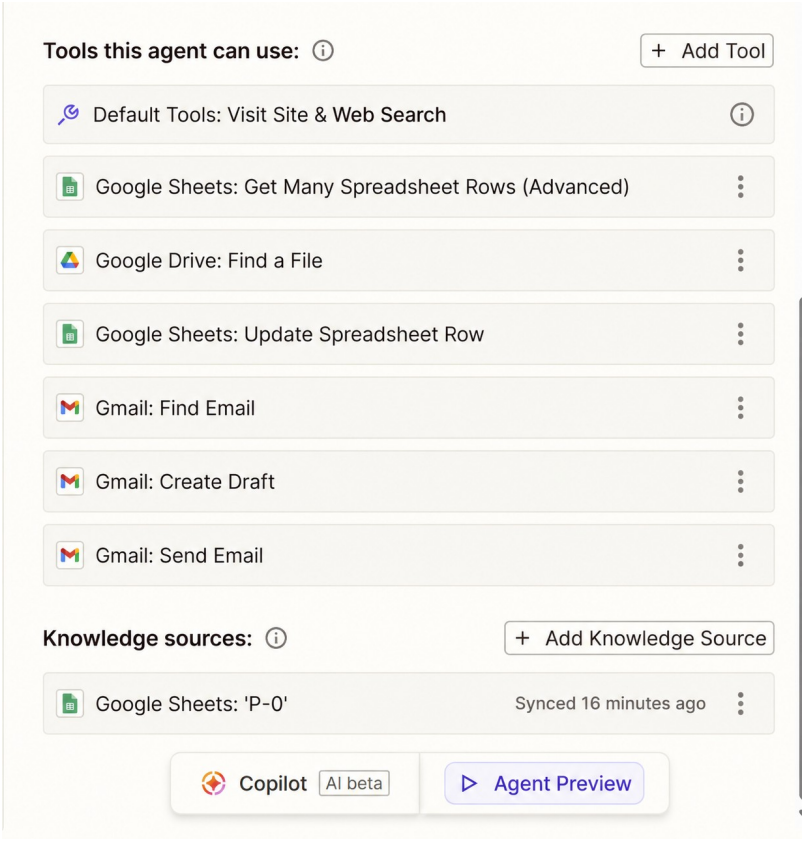


Figure 2: Tools and knowledge source connected to the agent

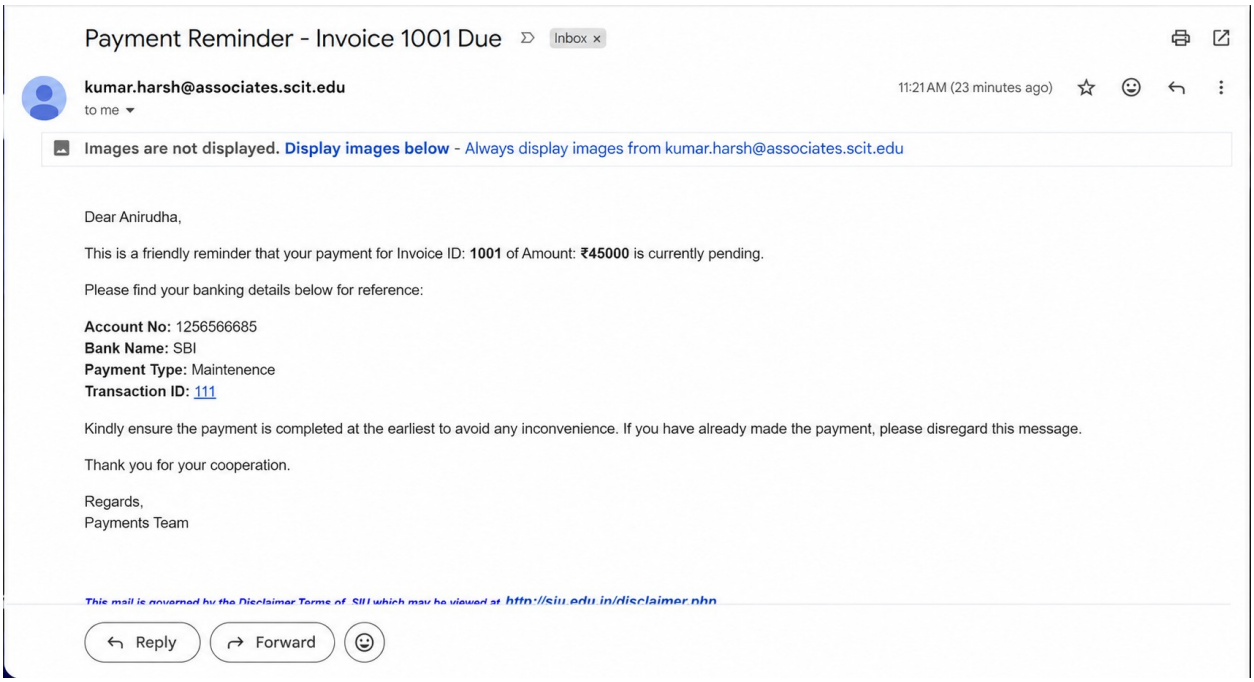
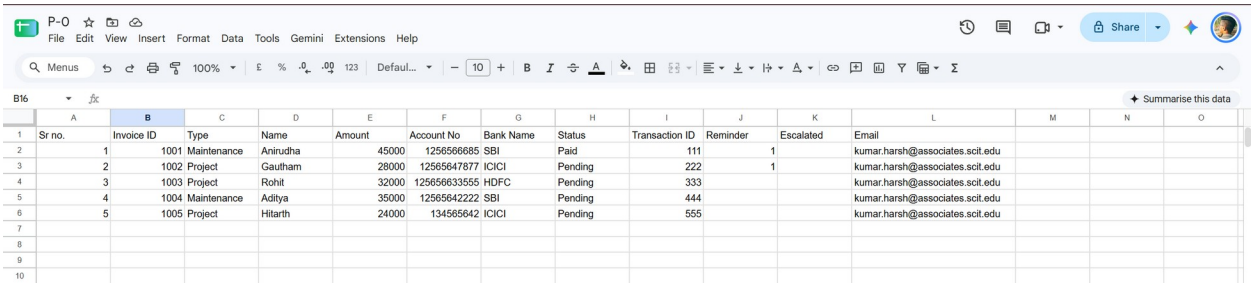


Figure 3: Payment reminder email received for Invoice 1001



P-0

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Summarise this data

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Sr no.	Invoice ID	Type	Name	Amount	Account No	Bank Name	Status	Transaction ID	Reminder	Escalated	Email			
2	1	1001	Maintenance	Anirudha	45000	1256566685	SBI	Paid	111	1		kumar.harsh@associates.scit.edu			
3	2	1002	Project	Gautham	28000	12565647877	ICICI	Pending	222	1		kumar.harsh@associates.scit.edu			
4	3	1003	Project	Rohit	32000	125656633555	HDFC	Pending	333			kumar.harsh@associates.scit.edu			
5	4	1004	Maintenance	Aditya	35000	12565642222	SBI	Pending	444			kumar.harsh@associates.scit.edu			
6	5	1005	Project	Hitarth	24000	134565642	ICICI	Pending	555			kumar.harsh@associates.scit.edu			
7															
8															
9															
10															

Figure 4: P-0 invoice spreadsheet with Reminder column updated

Note: *Designated test recipient email addresses are used for demonstration and validation of the automation. Sensitive credentials, passwords, API keys and private authentication information should not be shared or uploaded publicly.*